

API Security & Anomaly Inspection Report

Target Base URL:	http://127.0.0.1:5000
Session ID:	#17
Scan Date:	2026-09-17 13:24:56.045476
Overall Risk Score:	65.09 / 100
Overall Severity:	HIGH
Total Endpoints Tested:	16
Vulnerabilities Found:	2

Executive Summary

This inspection report summarizes telemetry security findings for **http://127.0.0.1:5000**. The scanner evaluated target endpoints across three independent detection layers: Layer 1 (Signature Rules), Layer 2 (Isolation Forest ML Anomaly), and Layer 3 (PyTorch LSTM + Autoencoder DL). Overall risk severity is assessed as **HIGH** with a composite score of **65.09/100**.

Vulnerability & Anomaly Findings

Endpoint	Attack Type	Severity	Score	Recommendation
http://127.0.0.1:5000/	SQL_Injection	HIGH	62.22	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/	XSS	HIGH	60.92	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/	Command_Injection	HIGH	59.14	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/	Security_Misconfiguration	MEDIUM	31.91	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/books/v1	SQL_Injection	HIGH	61.96	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1	XSS	HIGH	60.03	Significant anomaly or signature match detected. Review input validation and database query parameterization.

http://127.0.0.1:5000/books/v1	Command_Injection	HIGH	58.5	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1	Security_Misconfiguration	MEDIUM	31.66	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/books/v1/admin	SQL_Injection	HIGH	65.09	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/admin	XSS	HIGH	57.4	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/admin	Command_Injection	HIGH	57.4	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/admin	None	LOW	13.42	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.
http://127.0.0.1:5000/createdb	SQL_Injection	HIGH	58.73	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/createdb	XSS	HIGH	57.72	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/createdb	Command_Injection	HIGH	56.08	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/createdb	Verbose_Error_Exposure	HIGH	43.08	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/me	SQL_Injection	HIGH	60.27	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/me	XSS	HIGH	59.25	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/me	Command_Injection	HIGH	57.4	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/me	None	LOW	10.55	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.

http://127.0.0.1:5000/users/v1	Security_Misconfiguration	MEDIUM	31.66	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/users/v1/_debug	Sensitive_Data_Exposure	HIGH	50.43	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/users/v1/admin	Security_Misconfiguration	MEDIUM	35.62	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/users/v1/admin/email	None	LOW	13.31	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.
http://127.0.0.1:5000/users/v1/admin/password	None	LOW	13.31	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.
http://127.0.0.1:5000	SQL_Injection	HIGH	62.22	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000	XSS	HIGH	60.92	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000	Command_Injection	HIGH	59.14	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000	Security_Misconfiguration	MEDIUM	31.91	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/users/v1/name1	Security_Misconfiguration	MEDIUM	29.55	Elevated structural anomaly or header misconfiguration detected. Audit API endpoint input parameters.
http://127.0.0.1:5000/books/v1/1	SQL_Injection	HIGH	60.27	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/1	XSS	HIGH	58.09	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/1	Command_Injection	HIGH	56.67	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/1	None	LOW	10.48	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.

http://127.0.0.1:5000/books/v1/book1	SQL_Injection	HIGH	60.17	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/book1	XSS	HIGH	57.4	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/book1	Command_Injection	HIGH	57.17	Significant anomaly or signature match detected. Review input validation and database query parameterization.
http://127.0.0.1:5000/books/v1/book1	None	LOW	10.48	Baseline telemetry parameters normal. Maintain routine security monitoring and header hardening.

Detection Methodology

- 1. **Layer 1 (Signature Engine):** Performs deterministic regex matching on URLs, parameters, headers, and responses.
- 2. **Layer 2 (Machine Learning):** Analyzes the defined HTTP telemetry feature schema (17 features) using an Isolation Forest model trained on HTTP CSIC datasets.
- 3. **Layer 3 (Deep Learning):** Combines character-level LSTM sequence classification with PyTorch Autoencoder feature reconstruction error benchmarking.